HENRY CHEN

M.Sc. Materials Science | 3 Years' Professional Experience | B.Eng. Mechanical Engineering (In Progress)

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EDUCATION

B.Eng. Mechanical Engineering | OsloMet — Aug 2025 – Present

Specialization: Mechatronics

Relevant Coursework: Mechanical Design, Thermofluids, Python Programming, Statistics & Risk Analysis

M.Sc. Materials Science for Energy and Nanotechnology | University of Oslo — Aug 2017 – Jun 2022

- Synthesised proton-conducting ceramics by solid-state reactive sintering.
- Optimised precursors, milling, ZnO addition, and sintering conditions.
- Characterised microstructure, phase formation, hydration, and ionic transport using SEM-EDS, XRD, TGA, EIS, and EMF.

ENGINEERING SKILLS

Python & Data Analysis	OOP, Pandas, NumPy, Seaborn, Scikit-learn, Statsmodels
Simulation & Modelling	CFD (SimScale, OpenFOAM), Design of Experiments (DoE)
Design & Manufacturing	CAD (Inventor, Fusion 360, Onshape), 3D Printing and Laser Cutting (Makerspace OsloMet)
Materials Characterisation	SEM-EDS, XRD, HT-XRD, TGA, EIS, CV, EMF
Materials Processing	Powder Processing, Solid-State Synthesis, Reactive Sintering, APCVD

PROFESSIONAL EXPERIENCE

SEM Operator | ALS Laboratory Group Norway AS — May 2025 – Aug 2025

- Performed SEM-EDS analysis and reporting for asbestos identification.
- Operated and maintained SEM instrumentation.
- Prepared technical reports in accordance with accreditation requirements.

Research Engineer | N2 Applied AS — Apr 2023 – Mar 2025

- Investigated electrode degradation mechanisms using SEM-EDS characterisation.
- Performed CFD simulations of heat exchangers, cooling systems, and swirling flows.
- Designed and analysed experiments using DoE methodologies and Python.
- Applied statistical modelling to optimise plasma NOx generation processes.

Researcher | nanoCaps AS — Aug 2022 – Apr 2023

- Fabricated CNT-based CR2032 coin cells in a glovebox environment.
- Characterised electrochemical performance using CV and EIS.
- Synthesised carbon nanotubes using APCVD.
- Analysed CNT morphology using SEM imaging.

Board Member | Kjemisk Fagutvalg (KFU) ved Kjemisk Institutt - UiO — 2019

- Liaised between students and the Chemistry department to promote and safeguard student interests